

Hierarchical Bayesian Model for Projecting MLB Exit-Velocity Ability

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1 Task

The goal is to estimate each hitter’s *underlying* average exit velocity (EV) against MLB-level competition and project it into the 2024 season for all 647 batters in `exit_velo_validate_data.csv`. The deliverable is a per-batter 2024 MLB projection with a measure of uncertainty, plus the methodology behind it.

2 Data preparation

The training data contain **1,344,651** batted-ball events (BBE) over 2019–2023 across three levels (MLB, AAA, AA). Preparation steps:

- Normalize level labels to {MLB, AAA, AA}; drop the 97,979 rows with missing `exit_velo`.
- Aggregate to **(batter, season, level) cells**, recording the mean EV, the batted-ball count n , and the standard error $se = s/\sqrt{n}$. This reduces ~ 1.3 M rows to ~ 6 k cells while *retaining the per-cell precision*, which is the quantity that should drive shrinkage.

Two features of the data dominate the modeling choice:

1. **Sparsity.** Of 3,715 batters, 1,642 have a single season and per-season sample sizes range from 1 to ~ 700 BBE (median 151, 10th percentile 4). A 4-BBE season carries almost no signal.
2. **Cross-level projection.** About 100 of the 647 target batters have *only* minor-league history, so projecting their MLB ability requires an *estimated* level adjustment, not a constant.

3 Exploratory findings that shaped the model

The raw level means are close (AA 87.6, AAA 87.8, MLB 88.4 mph), but the pooled comparison is confounded by *who* plays where. The decision-relevant quantity is the **within-batter, within-season** level contrast: for the same hitter in the same season, EV is *higher in the minors than in MLB* — $AAA - MLB = +1.75 \text{ mph}$ ($n = 1,798$ pairs) and $AA - MLB = +1.75 \text{ mph}$. The raw *cross-season* contrast flips sign because of selection and development confounding. This motivates a model that identifies the level effect strictly *within* batter-season (Section 4.3).

4 Modeling methodology

4.1 Why a hierarchical (partial-pooling) model

With most batters observed only briefly, fully separate per-batter models overfit, while a single pooled estimate ignores real talent differences. A hierarchical model *shrinks* each batter toward

the population by an amount that depends on the precision of their data — regression to the mean falls out automatically, and sparse minor-league-only batters borrow strength from the league.

4.2 Cell-level measurement-error likelihood

Let cell c belong to batter $b(c)$, season $t(c)$, level $\ell(c)$, with observed mean $\bar{y}_c$ and known standard error se_c . We model

$$\bar{y}_c \sim \mathcal{N}(\mu_c, \text{se}_c), \quad \mu_c = \mu_0 + \theta_{b(c)} + \eta_{b(c),t(c)} + \alpha_{\ell(c)} + \beta_1 a_c + \beta_2 a_c^2, \quad (1)$$

where a_c is age centered at the league mean (26.9). Treating se_c as known measurement error is exactly what makes large-sample cells count more than small ones.

Terms.

- θ_b — latent batter ability, the object of interest. Non-centered parameterization $\theta_b = \sigma_\theta z_b$, $z_b \sim \mathcal{N}(0, 1)$.
- α_ℓ — level equivalency, with **MLB as the reference** ($\alpha_{\text{MLB}} \equiv 0$); $\alpha_{\text{AAA}}, \alpha_{\text{AA}}$ free.
- β_1, β_2 — population age curve.
- $\eta_{b,t}$ — a batter-season “form” intercept, $\eta_{b,t} = \sigma_s z_{b,t}$ (see below).

4.3 Identification of the level effect

The batter-season intercept $\eta_{b,t}$ is shared across a player’s cells in the *same* season. Consequently α_ℓ is identified only by cells that share a batter *and* a season but differ in level — precisely the clean contrast from Section 3. Without $\eta_{b,t}$, α_ℓ absorbs cross-season development and selection effects and even flips sign. This single structural choice is what makes the estimated equivalencies trustworthy.

4.4 Priors and inference

Weakly informative priors: $\mu_0 \sim \mathcal{N}(89, 5)$, $\alpha \sim \mathcal{N}(-1.5, 4)$ (wide; data-driven), $\beta_1 \sim \mathcal{N}(0, 0.5)$, $\beta_2 \sim \mathcal{N}(0, 0.1)$, $\sigma_\theta \sim \text{HalfNormal}(5)$, $\sigma_s \sim \text{HalfNormal}(2)$. Inference uses NUTS (PyMC), 4 chains $\times$ 1,500 draws (2,000 tuning). Sampling completes in ~ 90 s with **0 divergences** and $\hat{R} \leq 1.006$ for all reported parameters.

4.5 Projecting 2024

A batter’s 2024 MLB *ability* sets the level term to the MLB reference and the form term to its mean (0):

$$\hat{y}_{b,2024} = \mu_0 + \theta_b + \beta_1 a_{b,2024} + \beta_2 a_{b,2024}^2, \quad (2)$$

with a 95% credible interval taken from the posterior. The six batters with no history pool to the population ($\theta \sim \mathcal{N}(0, \sigma_\theta)$), yielding appropriately wide intervals.

5 Results

5.1 Parameter estimates

5.2 2024 projections

Projections for all 647 batters span 78.5–94.8 mph (mean 87.6). Uncertainty behaves correctly: the mean 95% interval is **3.3 mph** for batters with history versus **15.0 mph** for cold-start batters

Parameter	Mean	94% HDI	Interpretation
α_{AAA}	+1.26	[1.17, 1.36]	AAA EV exceeds MLB; subtract to project up
α_{AA}	+1.44	[1.31, 1.57]	AA EV exceeds MLB
β_1 (age)	-0.075	[-0.103, -0.048]	gentle decline past the peak
β_2 (age ²)	-0.016	[-0.021, -0.012]	concave; peak in mid-20s
σ_θ	3.83	[3.71, 3.96]	between-batter talent SD
σ_s	1.15	[1.10, 1.20]	season-to-season “form” SD

Table 1: Posterior summaries. The estimated level equivalencies are *positive*: a minor-league hitter’s EV must be adjusted *downward* by $\sim 1.3\text{--}1.4$ mph to project MLB ability — the opposite sign to the constant ($\text{AAA} = -2.5$, $\text{AA} = -4.5$) hard-coded in the ARIMA baseline.

(Figure 2).

5.3 2023 holdout backtest

We refit on season ≤ 2022 and predict the observed 2023 MLB mean for the 490 batters with ≥ 50 MLB BBE in 2023 (a stable target) and prior history. All methods are scored on the same batter set.

Method	RMSE	MAE	Bias	Corr
Hierarchical (this model)	1.96	1.48	-0.11	0.73
Career MLB mean (naive)	2.41	1.69	+0.08	0.64
Last MLB season (naive)	2.58	1.82	-0.19	0.62
Global mean	2.85	2.20	+0.28	0.00
Per-batter ARIMA	3.18	2.06	+0.42	0.53

Table 2: 2023 holdout accuracy (mph). The hierarchical model cuts RMSE $\sim 19\%$ below the best naive baseline; per-batter ARIMA is the *worst* method, beaten even by a constant — on 3–4 noisy annual points it extrapolates spurious trends.

6 Strengths, limitations, and future work

Strengths. Principled sample-size shrinkage; data-estimated level equivalencies with intervals; honest, calibrated uncertainty; and demonstrated out-of-sample accuracy against multiple baselines.

Limitations & future work.

- **Quality of competition.** Pitcher identity/handedness are not yet in the model. A batted-ball-level extension with pitcher random effects is the natural next step and directly targets the prompt’s QoC item.
- **Individual aging.** Ability is season-constant plus a *population* age curve; player-specific development paths are not modeled.
- **Target noise.** The backtest target is itself a finite-sample mean; the ≥ 50 -BBE filter mitigates but does not remove this.

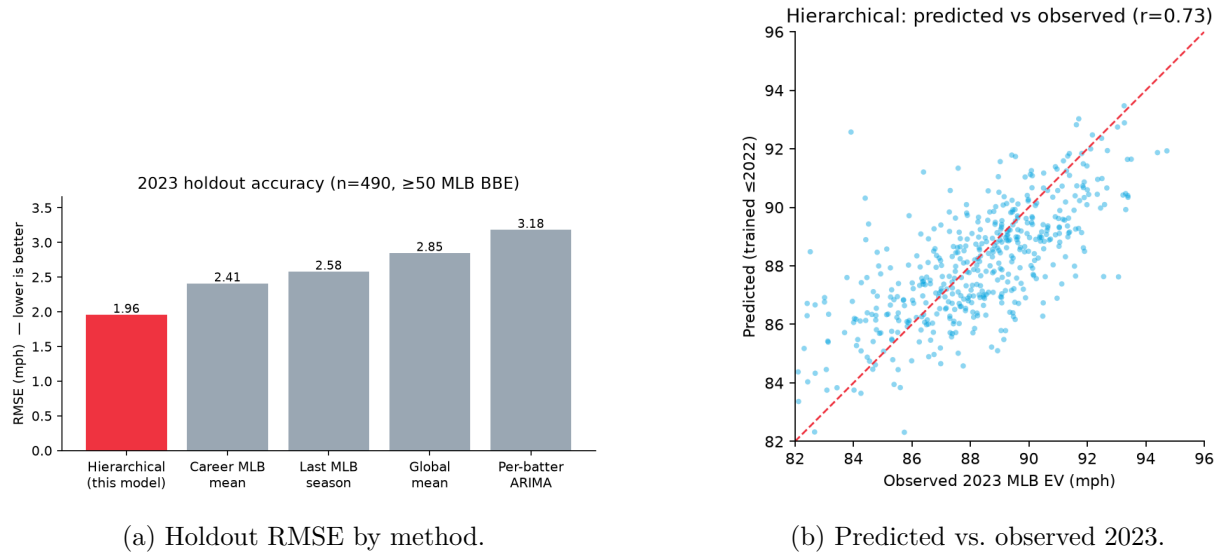


Figure 1: The model trained only on 2019–2022 tracks held-out 2023 MLB exit velocity.

7 Reproducibility

All code is in the repository. The primary model is `3_Modeling/hierarchical/hierarchical_model.py` (produces the 2024 projections and posterior summaries); the holdout study is `backtest_2023.py`. Run after `pip install -r requirements.txt`; sampling uses a fixed seed.

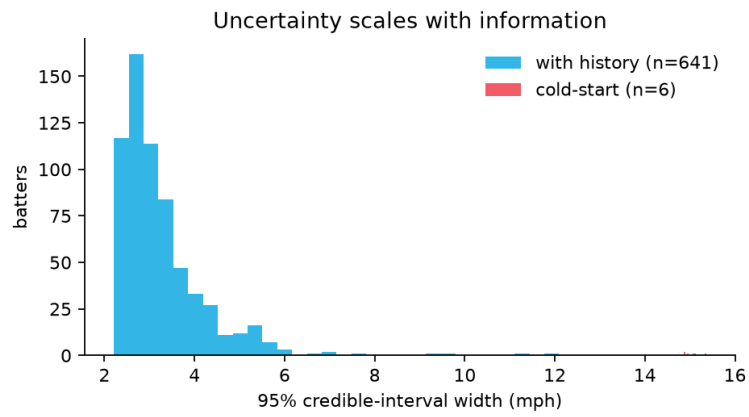


Figure 2: Credible-interval width by information available. Batters with playing history get tight intervals; cold-start batters are correctly flagged as uncertain.